STAT 330-001

40142034 37

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Entry: 01:18pm 10/21/2026 Checked out:

Employee: Louis R.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS

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037

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

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- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

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Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

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Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

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Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

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(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

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(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

The Coefficients of β_i are being Shrunk, It is accomplish by seeing which Predictor has low RMSE & Uses our λ to Strength Penalty to Shrink Coefficients

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

(Circle one answer) Which conclusion is most appropriate?

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Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\widehat{Y_i} = \widehat{B_0} + \widehat{B_1} X_i$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\widehat{B}$ is our fitted estimated values, while B is our original B w/ residual errors.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

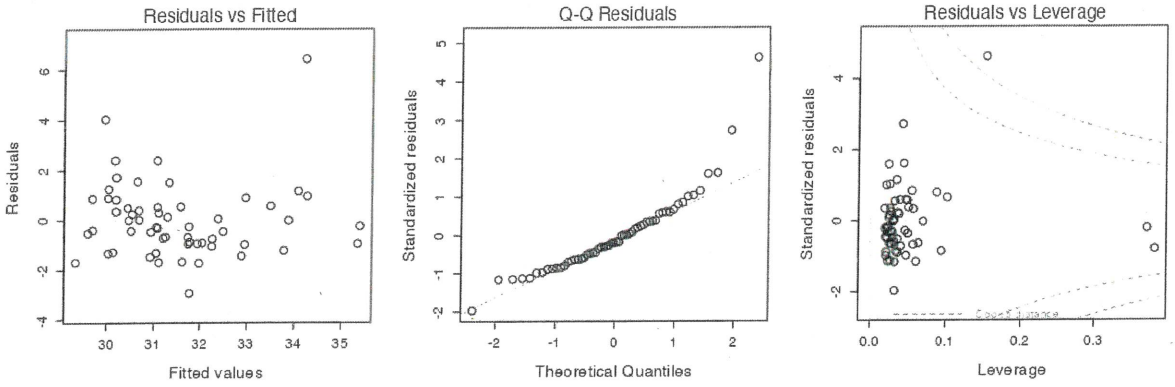
Our Least Squares looks at all our $\widehat{B}$ SSE. It then chooses the $\widehat{B}$ w/ the smallest SSE & that is our coefficient.

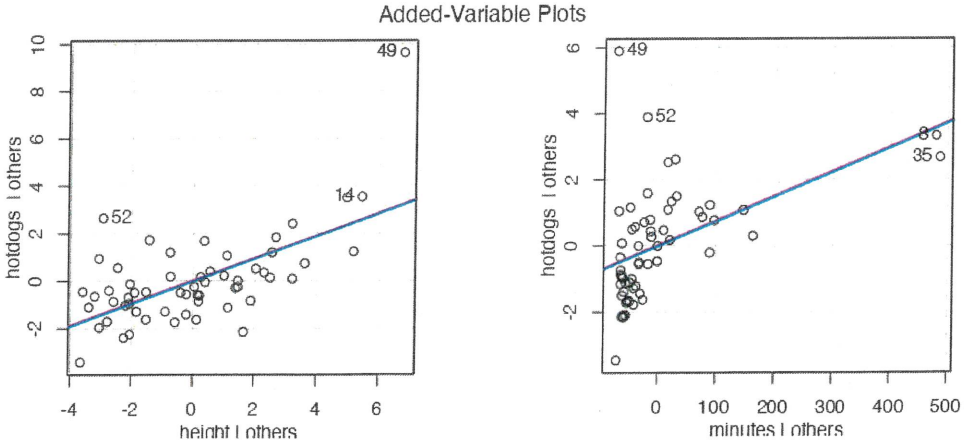
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	Pr($>\|t\|$)
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

$\beta_1 = .4728$ad, Holding minutes constant, a 1 inch increase in Height is associated w/ an estimated .4728x12 unit increase in avg # of Hotdogs Consumed in 10 min.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Looking at our Residual VS. Fitted Plot the points are evenly spread & look flat around 0. Also our added variable plots look linear

(B) There is independence between each observation

Each eater was timed seperately w/ there own hotdogs. One eater did not effect the results of another

(C) The errors are normally distributed

We can say it is Closely Normal Since our qq-plot the points follow the line, however, the tail ends do curve. Also our Shapiro-Wilk Pvalue is Very Small, So Normality is very hard to assume. for our data

(D) The errors have constant variance (homoscedasticity)

looking at our Resid vs. fitted the points are evenly spread & our Breusch-Pagan Pvalue is large So we can assume the errors have constant Variance

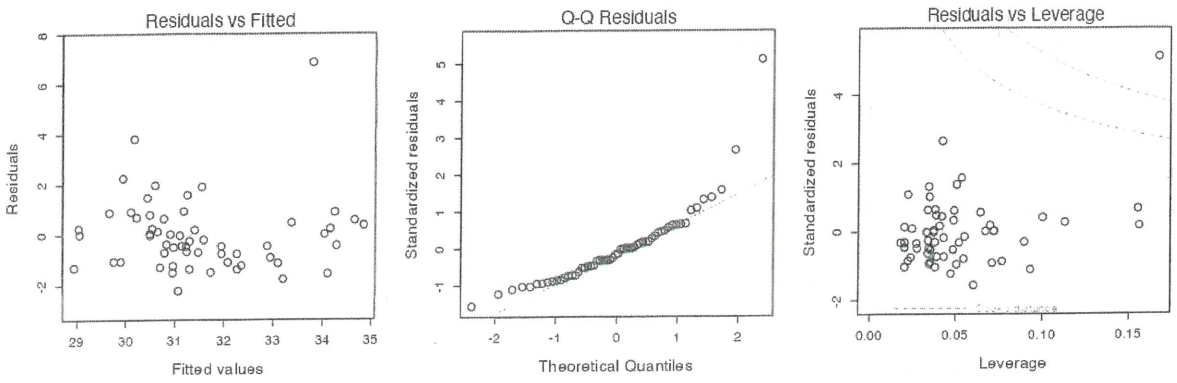
(E) There are no unduly influential points

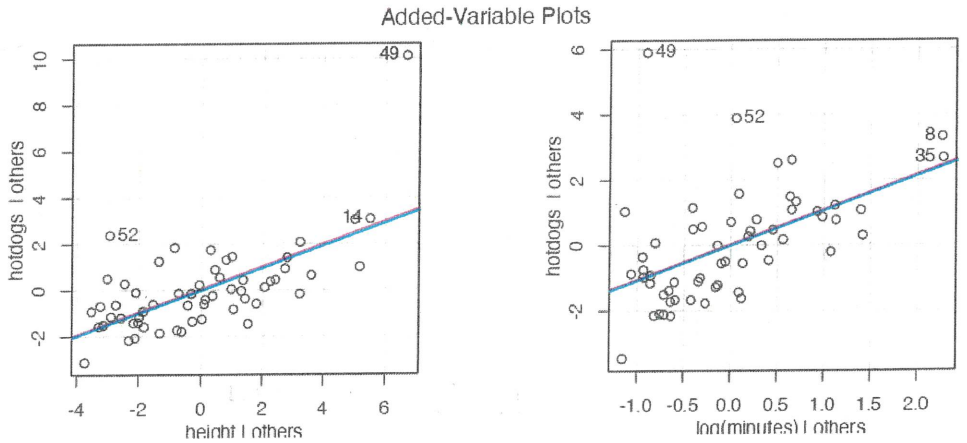
Looking at our Resid VS. Lev plot there is 1 point that is a outlier, so it should be investigated

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would say we can't directly say Model 1 is Superior. Although our model 2 homoscedasticity is hard to assume, there is other factors we must look at. I understand model 1 may be better since our E assumption is met but doesn't mean instantly is Superior

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Advantage: Minutes is fitted around a linear line in added-variable plot & our CV RMSE is smaller. Slightly more normal

Disadvantage: Breush-Pagan P-value is small, so Equal Variance is not met
- We have less influential Points from Resid vs. lev plot.

Question 16: State the estimated value of β_2 and provide an interpretation in context

$B_2 = 1.06702$, Holding Height Constant, a 1% increase in Minutes Spent training is associated w/ a 1.06702 Change in average # of Hotdogs consumed in 10 min.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$-6.05589 + .48834(68) + (1.06702)(4.369)$

$= 31.8134$

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55 $\quad 59 - 2 - 1$

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.8134 + 2.004(1.485) = 32.2214$

$[31.404134, 34.222866]$

$31.8130 - 2.004(.204)$

Part III - Application

Questions 20-24 concern the following dataset:

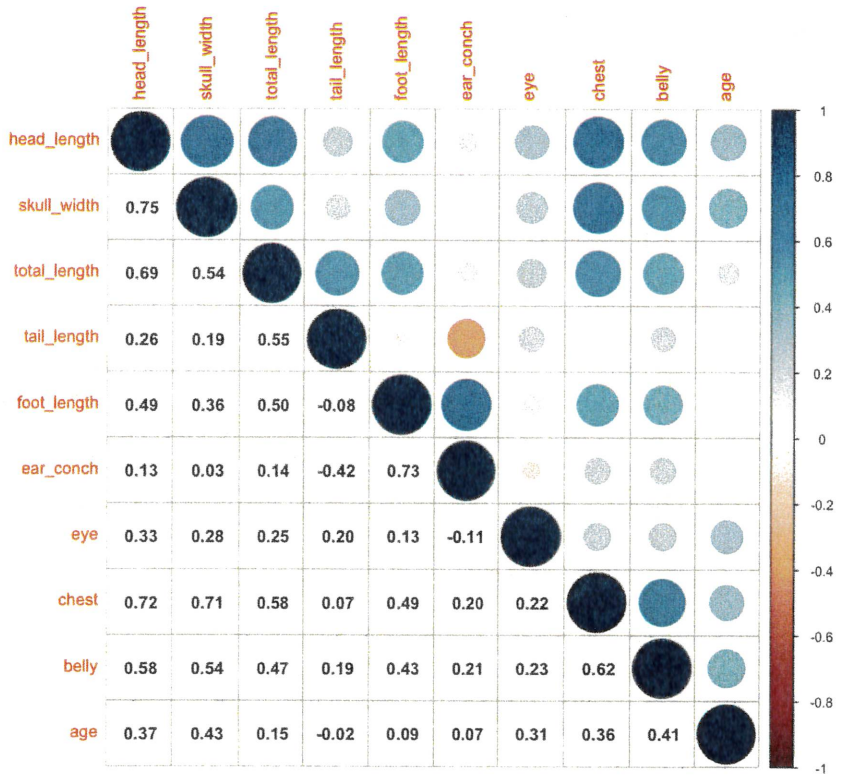
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

0.1655q + 2.01(.1059q) =

$0.1655q \pm 2.01(.1059q)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that our Coeffiant for Belly holding all other predicters constant is between [-.023, 0.443]

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

What is driving this is multicollinearity. Some predicters are correlated w/ others which results why each has no significant effect, but they affect each other & not just Age. Multicollinearity is our "Contradicter"

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Our best Subsets Selects which models we Should look at. And AIC is our metrix which See's each model & Chooses which will win or which is best based on what model gives lowest AIC.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

A Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 36

McCall, Peyton Anthon

739844778 peyton28 801-703-5591

Test: 639986 Form: A
Entry: 01:17pm 10/21/2026
Employee: Jessica Y.

Wadsworth, Spencer Gordon
Checked out:

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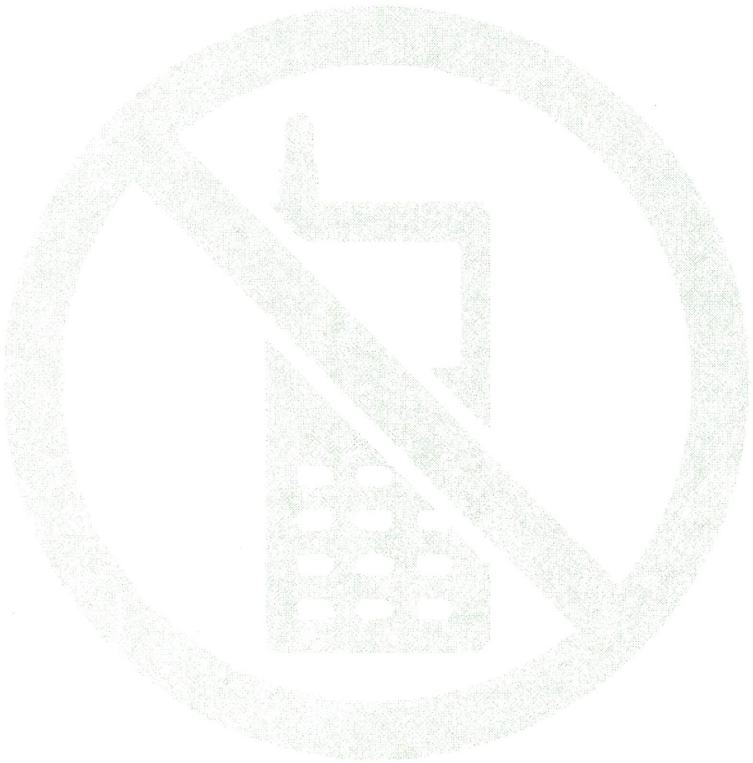

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(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Their formulas include a lambda(λ) which can shrink the coefficients(b). A higher lambda has a stronger penalty and will shrink the coefficients more.

$D_i > 0.04$

$D_i = \left(\frac{2.4}{4}\right)\left(\frac{0.25}{1-0.25}\right) = (0.6)(0.33) \approx 0.2$

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{4}{100} = 0.04$

(Circle one answer) Which conclusion is most appropriate?

(A) ~~$D_i = 0.48$.~~ Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) ~~$D_i = 0.48$~~. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

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$\hat{B} = y - xB$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{B}$ is our predicted value and B is our actual value.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

OLS chooses the smallest b values. Those b values are used in RSS, so smaller b values will make our RSS smaller.

PART II

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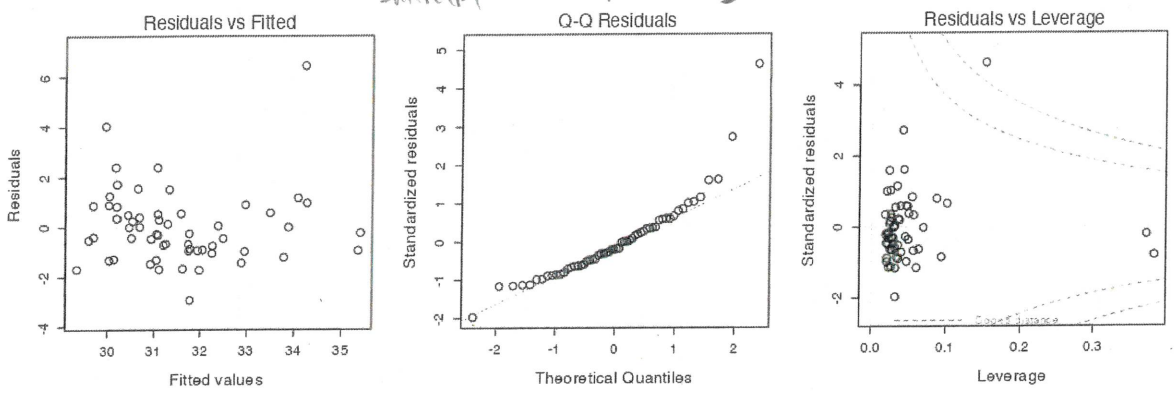
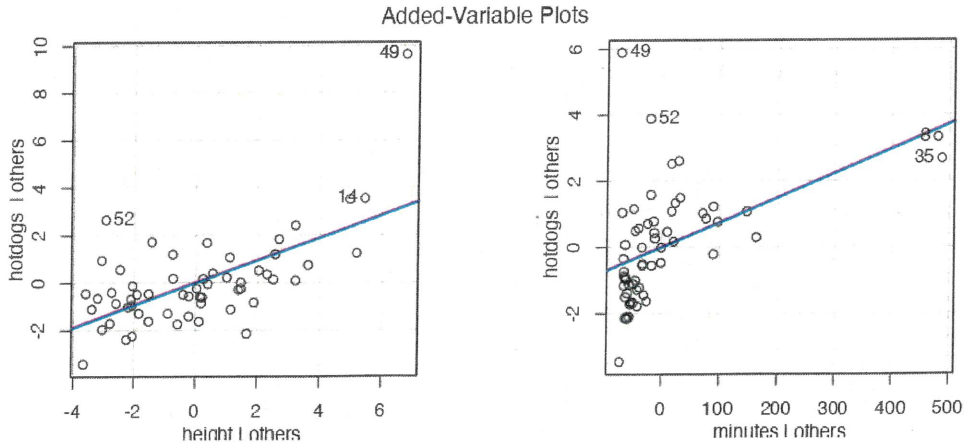
Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

Holding minutes constant, a one inch increase in height is associated with a 0.472892 increase in hot dogs eaten.

Question 12: **(Circle all that apply)** Which of the following are true?

$\beta_2 \pm t * SE(\beta_2)$

$0.004417 \pm 3.963(0.001871)$

$0.00002227, 0.00831773$

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The scattering on the residuals vs. fitted plot looks good, however, there is some curvature which could suggest some non-linearity.

(B) There is independence between each observation

Based on how the data was collected and observed, I see no issues with indeandence and all assumptions are meant.

(C) The errors are normally distributed

The Q-Q plot has some tail bending which could suggest innormality, and our Shapiro p-value is < 0.05, so we reject the null hypothesis and conclude that there could be innormality.

(D) The errors have constant variance (homoscedasticity)

The spread in the residuals vs. leverage plot is bunched and causes concern for inequality of constant variance, however, the Breusch test's p-value is > 0.05, so we fail to reject the null and suggest that errors have constant variance.

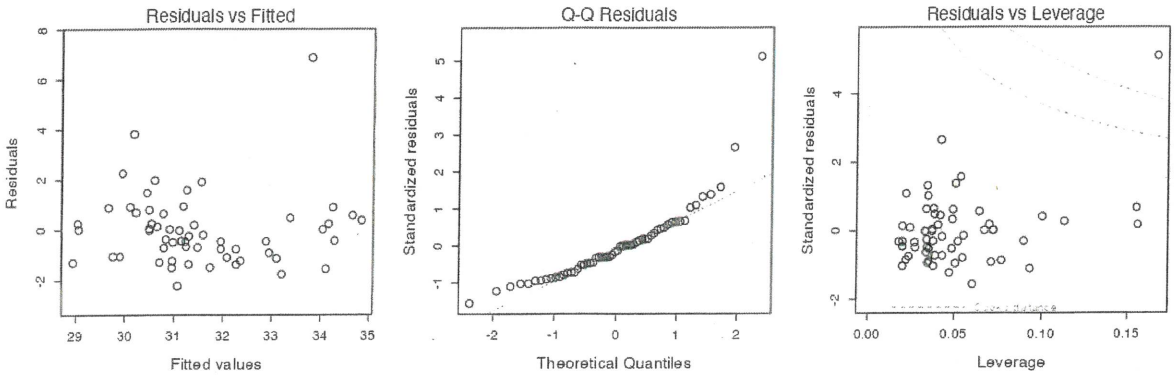
(E) There are no unduly influential points

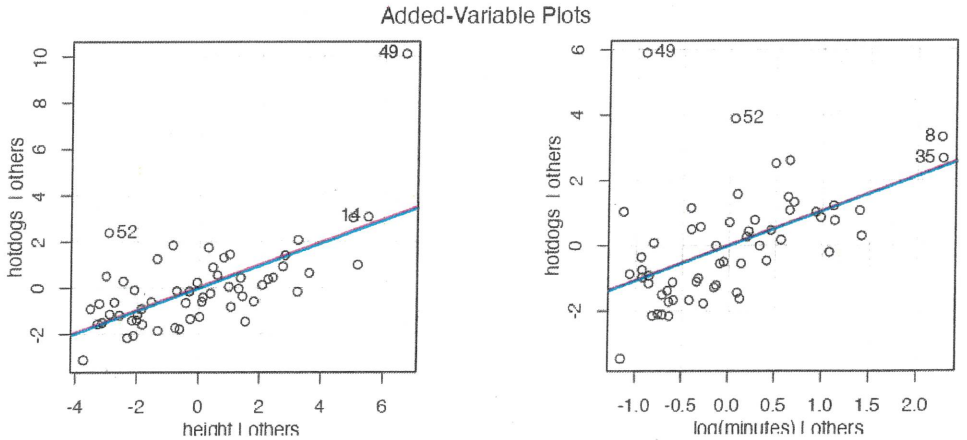
In both added variable plots, observation 419 does seem undly, so this observation should be investigated for errors and context.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I wouldn't conclude that model 1 is superior based only off the Breusch-Pagan test. Yes, it is an advantage of model 1 that the Breusch-Pagan test is bigger, but we also need to look at all other assumptions and tests between the two.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

The CV RMSE is slight lower in model 2, which is an advantage, along with the residuals vs. leverage plot looking more spread, and the Residuals vs. fitted looking flatter. However, Model has disadvantages with the Breusch-Pagan test being lower, there is still a tail bend in the Q-Q plot, and observation 49 is still a concern to this model

$\frac{1.06702}{100} = 0.0106702$

Question 16: State the estimated value of β_2 and provide an interpretation in context

Holding height constant, a 1% increase in log(minutes) is associated with a 0.0106702 increase in hot dogs eaten.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

hotdogs = -6.05589 + 0.48834(68) + 1.06702(4.369)

hotdogs = -6.05589 + 33.20712 + 4.66181038

hotdogs = 31.81304038

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

n-p-1

58-2-1

55

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$PI = \hat{y}_0 \pm t^* \sqrt{SE(\hat{y}_0)^2 + s^2}$

$PI = 31.81304038 \pm 2.004 * \sqrt{(0.204)^2 + (1.485)^2}$

$PI = 31.81304038 \pm 2.004 * \sqrt{0.041616 + 2.205225}$

$PI = 31.81304038 \pm 2.004 * \sqrt{2.246841}$

$PI = 31.81304038 \pm 2.004 (1.498947663)$

$PI = 31.81304038 \pm 3.003889047$

PI = (28.80915133, 34.81692943)

Part III - Application

Questions 20-24 concern the following dataset:

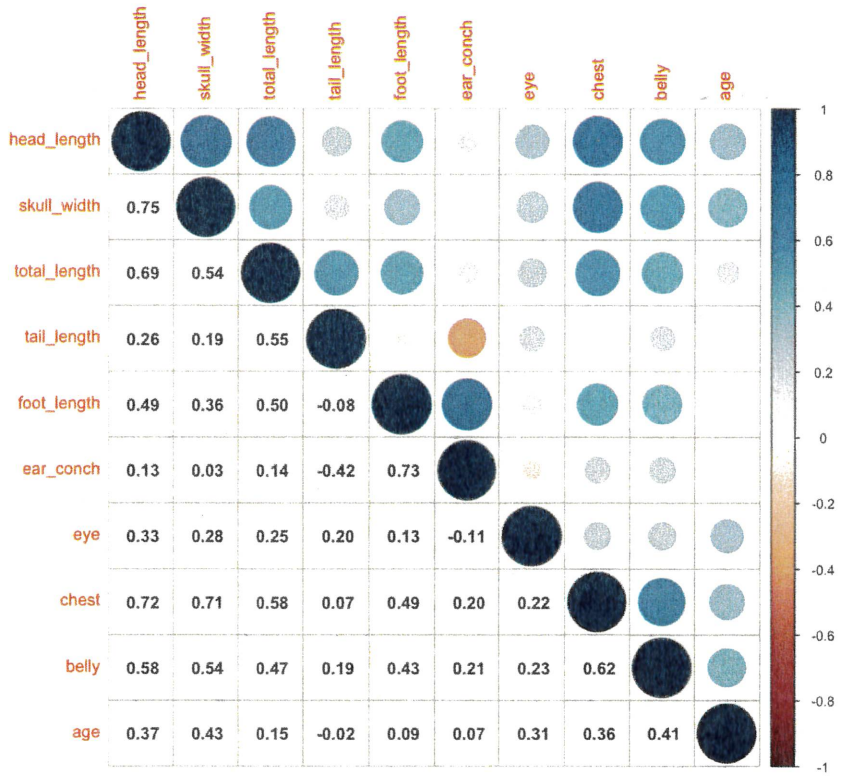
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -13.22377    7.34105  -1.801   0.0778 .
head_length    0.07077    0.10120   0.699   0.4877
skull_width    0.16559    0.10599   1.562   0.1247
total_length  -0.04752    0.10167  -0.467   0.6423
tail_length   -0.05049    0.17030  -0.296   0.7681
foot_length   -0.21499    0.10733  -2.003   0.0507 .
ear_conch      0.15730    0.10231   1.538   0.1306
eye            0.43736    0.22549   1.940   0.0582 .
chest          0.01883    0.18154   0.104   0.9178
belly          0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 2.01(0.10599)$

$(-0.0474499, 0.3786299)$

$0.16559 \pm 2.01(0.10599)$

$(-0.0474499, 0.3786299)$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We are 95% confident that the coefficient on belly-width is in the interval $(-0.023, 0.443)$, holding all else constant.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

The p-value of our test statistic is less than alpha, so we reject the null hypothesis and assume that at least one coefficient has a non-zero slope.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

Best subsets is a method-search algorithm that tests every combination of coefficients to find the best one. AIC is a method-selection that best subsets uses as criteria to select the best combination of coefficients for our model.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

C Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

C Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

C Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 35

Mendez, Aidee Deyanira

225352326 mendez13 918-804-1057

Test: 639862 Form: A
Entry: 12:42pm 10/21/2026
Employee: Jessica Y.

Wadsworth, Spencer Gordon
Checked out:

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS

40142034

035

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

Lasso & ridge are often referred to as "shrinkage" because Lasso can reduce coefficients close to 0 & ridge toward 0. This is accomplished by Lasso doing the absolute value of coefficients & ridge the squared coefficients

$\left(\frac{(2.4)^2}{3+1}\right)\left(\frac{.25}{1-.25}\right)$

$(1.44)\left(\frac{.25}{.75}\right) = .48$

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{4}{100} = .04$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$\hat{B} = B^T (X^T X)^{-1} X^T y$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{B}$ = sample estimate / part

B = parameter of samble / whole

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

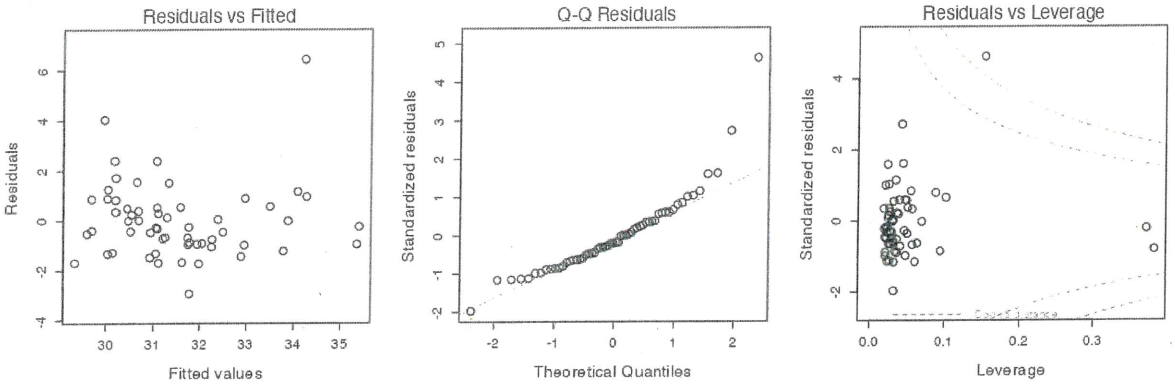
Least squares can be used to estimate regression coefficients by analyzing the data from the parameter, than using the sample estimate to find its correlation to the data.

PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$



Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

The estimated value of β_1 is .472842. holding the minutes per week each contestant spent per week training constant, each x, height increases by .472897

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0 $.4729 \pm (1.998)(.080352)$ $.31259$

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

There is a linear pattern between explandtory + response variable based on the graph.

(B) There is independence between each observation

Independence is affected by how the data was collected. The data was collected from a specific category, amateurs, which helps with independence

(C) The errors are normally distributed

$p < .05$. There is evidence to support that the distribution is not normally distributed because shapiro wilk value $< .05$.

(D) The errors have constant variance (homoscedasticity)

$p > .05$. The pagan value is greater than .05, which leads to not enough evidence to support homoscedasticity. Confidence interval, hypothesis may be unreliable

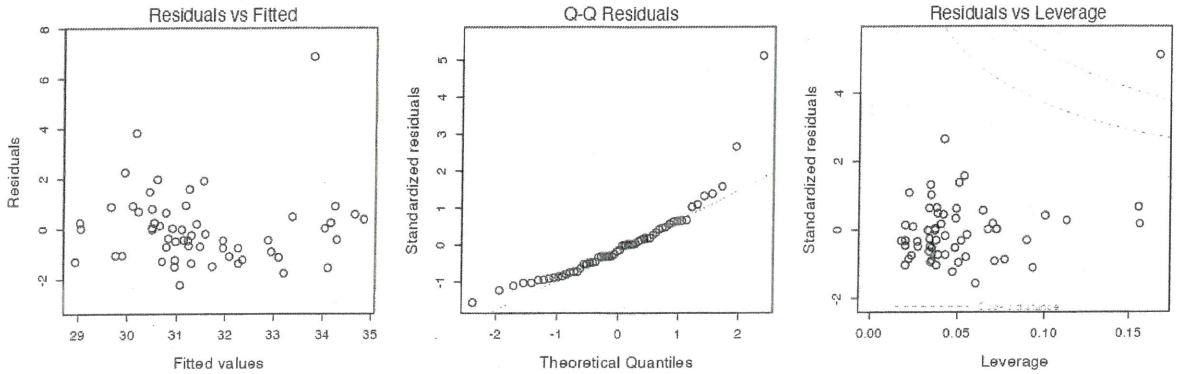
(E) There are no unduly influential points

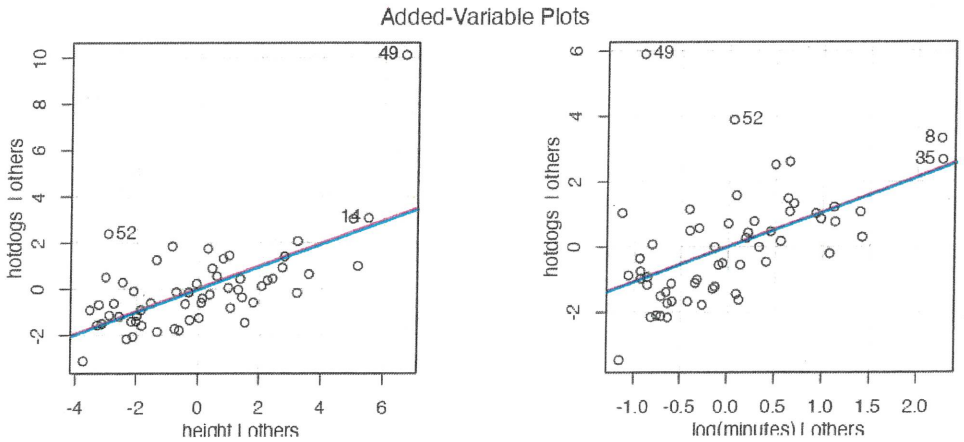
We have a low CV which indicates better prediction.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
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Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would respond that when conducting a test model & The answer in some areas differ, it tells us where is worth noting & revising our work rather than tell us that a complete model should be thrown out - That is why we look at other aspects such as CV RMSE, Shapiro Wilk, & ETC.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Some of the advantages of model 2 is that it was log transformed & a disadvantage is that it has been modified which can alter some of the data.

Question 16: State the estimated value of β_2 and provide an interpretation in context

$B_2 = 1.06702$. holding x_1 height constant, x_2 log minutes that a contestant spends per week traning per week increases by 1.06702.

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that $\log(79) = 4.369$. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

68 inch tall
79 minutes per week
log(79) = 4.369

$e^{4.369} = 78.964$
$= 79/10 = 7.8964$

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

$df = n - p$

(A) 55

(B) 57

(C) 56

(D) 48

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$\hat{B} \pm t^* SE(B)$

$(7.896) \pm (2.004)(1.485)$

7.896 ± 2.97594

$4.9200, 10.87194$

Part III - Application

Questions 20-24 concern the following dataset:

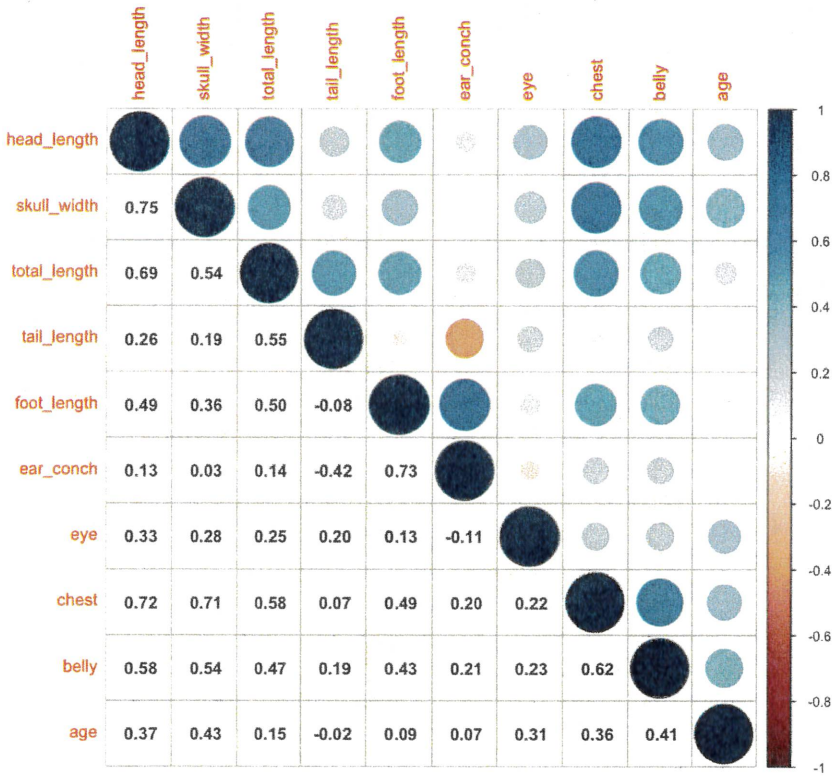
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

.16559 ± (1.562)(.10599)

.16559 ± (1.562)(.10599)

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

We fail to reject H_0. There is not enough evidence to support that $\beta = 0$ differs from 0.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

This contradiction supports why we fail to reject H_0 because the evidence supports normality, variance, etc.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

It performs variable selection by choosing random variables among the possible subsets.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

B Average the predictions and record their value

C Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

C Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

B Compute each model's AIC and BIC

A Fit each model using the testing data

A Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

B Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 34

Kelley, Korbin

997920466 korbinmk 702-927-8579

Test: 639771 Form: A
Entry: 12:12pm 10/21/2026
Employee: Jessica Y.

Wadsworth, Spencer Gordon
Checked out:

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly. Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

NS

40142034

034

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

these methods shrink the coefficient estimates by favoring bias to decrease variance.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right). \qquad D_i = \frac{2.4^2}{4} \cdot \frac{0.25}{0.75} = \frac{2.4^2}{12}$$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\hat{\beta}$?

$$\hat{\beta} = (X^T X)^{-1} X^T Y$$

(B) What is the conceptual difference between $\hat{\beta}$ and β?

β is a vector of fixed values that are associated with the population, and $\hat{\beta}$ are estimates of those values that come from a specific sample.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

Least squares estimates $\hat{\beta}$ by finding values that minimize the sum of errors.

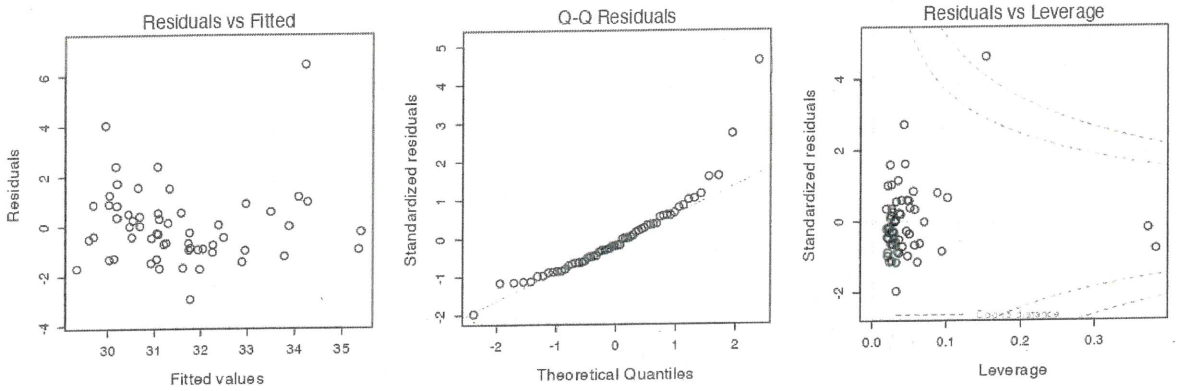
We can find $SSE(\beta)$ with the formula $(Y - X\beta)^T(Y - X\beta)$ which is the same as squaring each error and summing it (Y is observed, $X\beta$ is predicted). By expanding this $SSE(\beta)$ formula and simplifying, we can solve for the $\hat{\beta}$ estimates. $\left(\hat{\beta} = (X^T X)^{-1} X^T Y\right)$.

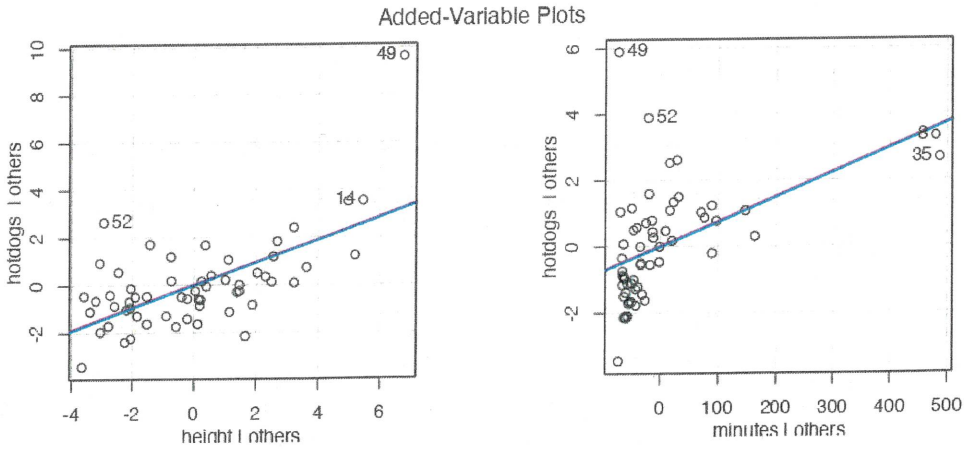
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

$0.007417 \pm 3.963 \cdot 0.001871$

0.007417 ± 0.007415

~~(0.000002, 0.014832)~~ $(0.000002, 0.014832)$

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

β_1 is approximately 0.4729

Holding all other predictors constant, a 1-inch increase in participant's height is associated with a 0.4729 increase in hotdogs eaten on average.

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0 0.00

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

The residuals vs. fitted values plot shows that the points are mostly centered on zero. Looking at the added-variable plots, there is some curvature on the minutes plot. Linearity is not perfect, but we don't have enough evidence to reject this model based solely on the linearity assumption.

(B) There is independence between each observation

From what we know about the data, we should be able to assume independence. If two or more competitive eaters that are part of the sample are friends and train together, it could cause some independence problems. But from what we know, this assumption is fine.

(C) The errors are normally distributed

The Q-Q plot has a bit of skew in the tails, and the Shapiro-Wilk p-value is very small. This suggests that the distribution is not normal. We shouldn't automatically discard our model, but we should investigate closer.

(D) The errors have constant variance (homoscedasticity)

The residuals vs fitted plot has a slight funnel shape, and the Breusch-Pagan p-value is 0.071. This isn't significantly small ($p<0.05$), but it suggests that the variances have some homoscedasticity. Not enough to make much of a fuss about.

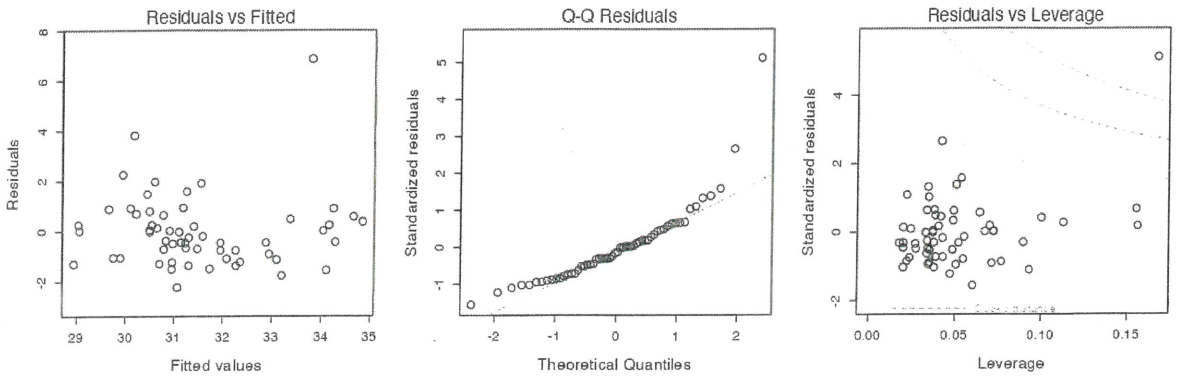
(E) There are no unduly influential points

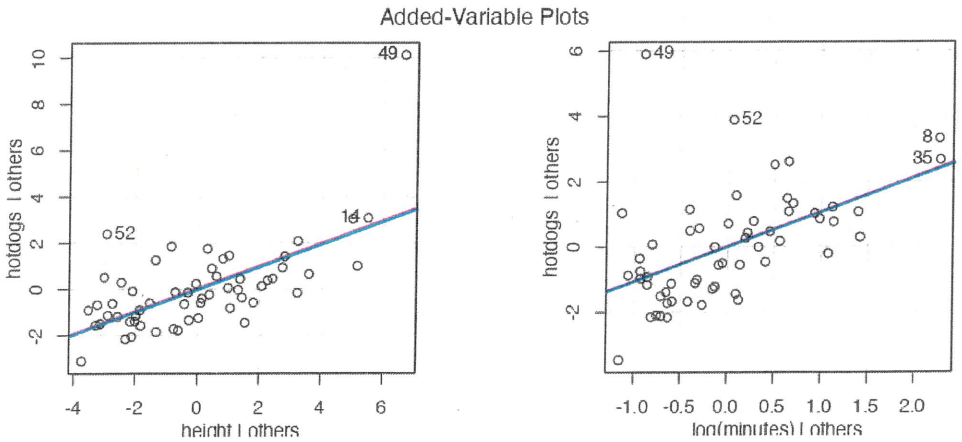
By looking at the residuals vs leverage plot, we see that there is one point with a high Cook's distance. We shouldn't immediately discard it, but we should investigate that point and see what might be happening with it.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

I would disagree. Model 1 may be favorable in that regard, but model(2) does ~~[illegible]~~ better with linearity. We can't assume one model to be superior from only one assumption. Because each model has parts that are preferable, we should use both and compare further to see what each one may be good for. (prediction, fit, etc.)

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

Model 2 has better linearity than model one; we no longer see the curvature in the added-variable plots. It also has a smaller CV RMSE suggesting it may be better for prediction. However, it fails both the Breush-Pagan and Shapiro-Wilk tests, meaning model 1 may be better as far as our assumptions go.

Question 16: State the estimated value of β_2 and provide an interpretation in context

β_2 is approximately 1.067

Holding all other predictors constant, a 10% increase in minutes trained gives ~~an~~ a 0.0442 unit increase in mean hot dogs eaten.

$1.067 \cdot \log(1.1)$

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that $\log(79) = 4.369$. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$\text{hotdogs} = \beta_0 + \beta_1 \text{height} + \beta_2 \log(\text{minutes})$

$= -6.05559 + 0.48834 \cdot 68 + 1.06702 \cdot 4.369$

$= 31.813$ hotdogs

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$DF = n - p - 1$

$= 58 - 2 - 1$

$= 55$

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

$31.813 \pm 2.004 \cdot 1.485$

$= 31.813 \pm 3.0294$

$(28.7836, 34.8424)$

Part III - Application

Questions 20-24 concern the following dataset:

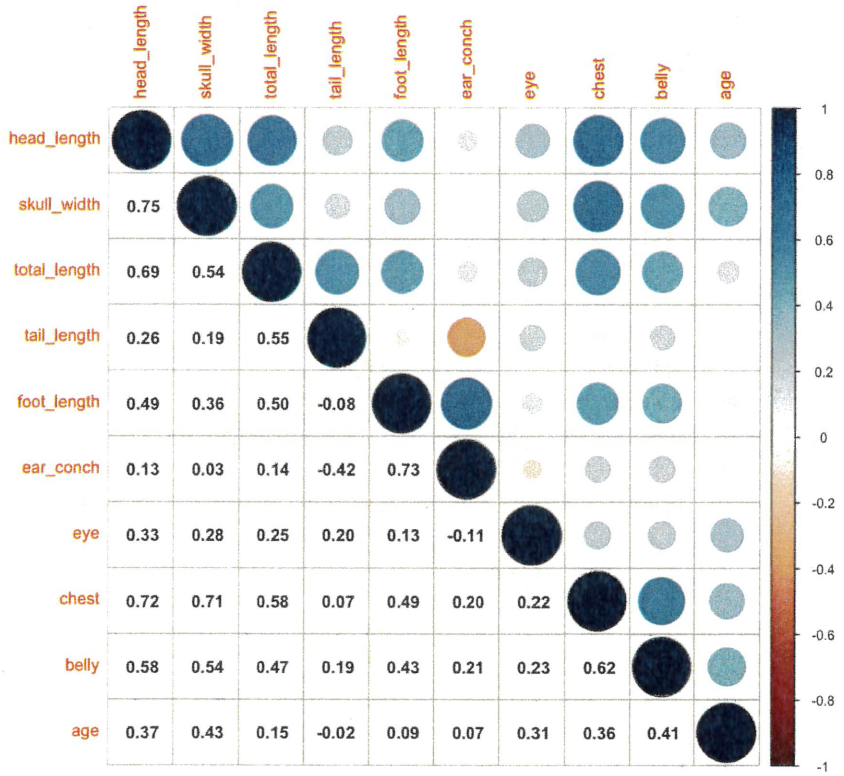
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

$0.16559 \pm 2.01 \cdot 0.10099$

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

~~We are 95% confident that the linear coefficient for~~

Holding all other predictors constant, we are 95% confident that a 1 cm increase in belly girth is associated with a change of between -0.023 an 0.0443 years of age of a possum on average

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

This could be a result of multicollinearity. If two or more factors are highly correlated, their individual t-tests may not result in a very small p-value even though our F-test does. If we try different models to minimize multicollinearity, we could see different results.

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The best subsets algorithm using AIC as its criterion goes through all 2^p possible models and compares the resulting AIC, and then gives us the model that it found to have the smallest AIC.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

D Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

A Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM

STAT 330-001

40142034 33

Tanner, Salinda Janae

305877262 salindaj 480-406-3853

Test: 639767 Form: A Wadsworth, Spencer Gordon

Entry: 12:11pm 10/21/2026 Checked out:

Employee: Jessica Y.

SPECIAL INSTRUCTIONS

TC Calc
Notes: 3x5
Dictionary
Take a deep breath. Be careful to read instructions thoroughly.
Answer questions completely.

Please keep this sheet **VISIBLE** on your desk at all times. When you are finished, *staple this sheet* to the front of your test booklet and return all materials to the Grading Table. You may use the space below for notes or scratch paper.

PLEASE TURN OFF YOUR CELL PHONE

40142034

033

STAT 330 - Exam 1

Section 001 (Wadsworth), Section 002 (White)
October 16th - 21st, 2026

Instructions

- You may not refer to your notes, other people, or other resources. You can, however, use one 3 inch by 5 inch note card that you create (writing on both sides is allowed).
- Do not discuss any aspect of this exam with anyone except a Stat 330 TA or instructor until scores are posted.
- You are allowed to use a calculator for this exam.
- In Parts II and III, **make sure to clearly show your work for any calculations that you make and to give your answers in context of the data sets**. Place your final answer inside the box, if one is provided.
- There are 24 questions. Make sure you answer all of them.
- There is extra scratch paper at the end of the exam.

PART I

Question 1: (Circle one answer) Partial regression (or added-variable) plots illustrate the effect of adding an additional variable to the model after accounting for the variables that are already included in the model.

(A) TRUE

(B) FALSE

Question 2: (TRUE/FALSE) If the relationship between x and y is nonlinear, a simple linear regression model may still be feasible after transforming the data.

- TRUE
- FALSE

Question 3: (Circle one answer) When adding an additional variable to a linear regression model, the model's R^2 ...

(A) always increases or remains unchanged.

(B) always decreases or remains unchanged.

(C) may increase or decrease depending of the additional variable.

Question 4: (Circle all that apply) Which of the following statements describe possible effects of multicollinearity?

(A) It can cause the linearity assumption to be violated.

(B) It can cause estimated coefficients to have signs opposite those of their true population coefficients.

(C) It can result in larger standard errors, and thus greater uncertainty in estimates.

(D) It can cause the coefficient of determination (R^2) to become negative.

Question 5: (Circle all that apply) Which of the following are true statements about multiple linear regression (MLR)?

(A) BIC will usually favor a model with more predictors than the AIC criterion when performing model selection.

(B) Cook's distance can be used to identify influential observations.

(C) Transformations of the x's or y may help alleviate problems with linearity, equal variance, or normality.

(D) The least-squares estimate of β can be obtained by fitting a simple linear regression for each x_k only.

(E) The assumptions on the error variables (i.e., the ϵ_i) in a multiple linear regression model are exactly the same as they are for simple linear regression.

Question 6: (Circle one answer) What is the primary benefit of using cross-validation (CV)?

(A) CV reduces variance in the sampling distribution of regression coefficients by splitting the training and test data.

(B) CV gives a more realistic representation of model performance by predicting observations that were not used to fit the model.

(C) CV ensures model assumptions are met which eliminates a time-consuming step of the modeling process.

(D) CV is the most efficient way to calculate the shrunken coefficient estimates in LASSO and ridge regression models.

Question 7: (Circle all that apply) In datasets with high multicollinearity and/or redundant covariates, what advantages do LASSO and/or ridge regression have in comparison to the traditional ordinary least squares (OLS) approach to model fitting?

(A) LASSO can set coefficients to be exactly equal to zero, and thus serves as an automated approach to variable selection

(B) Their coefficient estimates are less biased than those of OLS

(C) They produce more accurate confidence intervals and hypothesis tests

(D) Ridge regression often handles multicollinearity well

(E) They can handle datasets where the number of predictors exceeds the number of observations

(F) They can take advantage of the bias-variance tradeoff to produce more stable and accurate predictions

Question 8: Why are LASSO and ridge regression sometimes referred to as "shrinkage" methods? Specifically, what is being shrunk and how is this accomplished?

These are called shrinkage methods because they "shrink" the amount of variance found in the model by introducing some bias.

Question 9: A multiple linear regression model with $p = 3$ predictors is fit using $n = 100$ observations. Observation i has standardized residual $r_i = 2.4$ and leverage $h_{ii} = 0.25$.

Use

$$D_i = \frac{r_i^2}{p+1}\left(\frac{h_{ii}}{1-h_{ii}}\right).$$

$\frac{2.4^2}{4}\left(\frac{.25}{.75}\right)$

$1.44(.333) = .48$

(Circle one answer) Which conclusion is most appropriate?

(A) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(B) $D_i = 0.48$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(C) $D_i = 0.48$. The observation is not influential because its leverage is less than 0.50.

(D) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation is potentially influential and should be investigated.

(E) $D_i = 0.2$. Because this exceeds $4/n = 0.04$, the observation must be deleted before reporting the model.

(F) $D_i = 0.2$. The observation is not influential because its leverage is less than 0.50.

(G) Cook's distance cannot be calculated without knowing the observation's original residual.

Question 10:

(A) What is the formula for $\widehat{\beta}$?

$Y = X\hat{\beta} - e$

(B) What is the conceptual difference between $\widehat{\beta}$ and β?

$\hat{\beta}$ is the estimate found using the sample data. This will change depending on the sample.

β is the parameter. This is fixed for the population.

(C) Explain how least squares can be used to estimate regression coefficients. You do not need to derive anything, but you should be clear about how this is done conceptually.

$SSE(\beta) = (y - X\beta)^T(y - X\beta)$

if you take the derivative

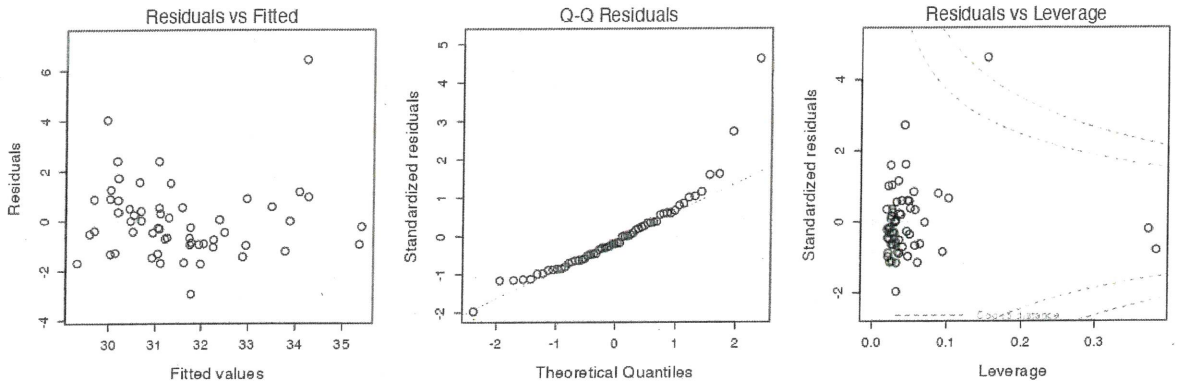
$\beta = (X^TX)^{-1}X^Ty$

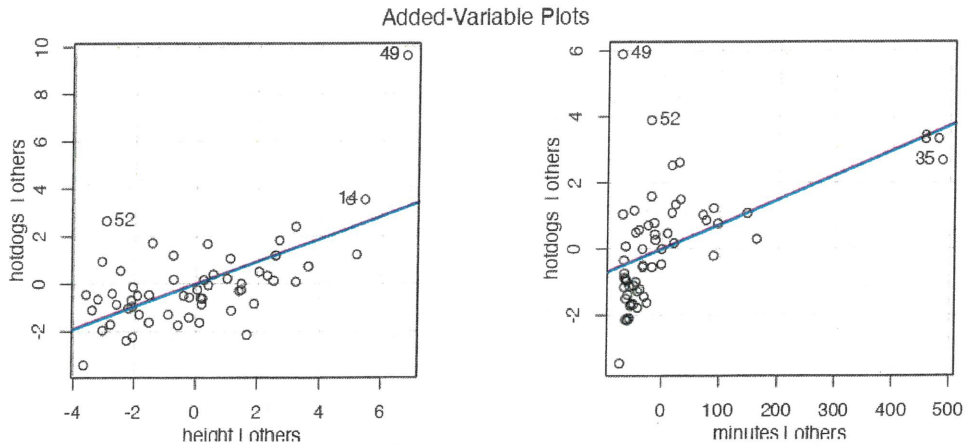
PART II

Some consider competitive eaters to be among the most inspiring individuals in modern times. We are interested in knowing what factors relate to a competitive eater's ability to consume. In a recent state fair, a contest was held where contestants competed to see who could eat the most hotdogs in 10 minutes. Data were collected on 58 amateur competitive eaters, most of whom were from the 4 largest counties in the state. For each eater, the total number of hotdogs consumed in a single 10 minute sitting was recorded. Each contestant's height in inches was also recorded, and each contestant was asked "How many minutes per week do you spend training for a contest?" The variables are `hotdogs`, `height`, and `minutes`. Model (1) is used for analysis. Use the plots and model fit output below to answer questions 11-13.

Model 1

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \texttt{minutes}_i + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{1}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-1.295392	5.467711	-0.237	0.813602
`height`	0.472892	0.080352	5.885	2.46e-07
`minutes`	0.007417	0.001871	3.963	0.000215

Other Summaries	Value
Breusch-Pagan p-value	0.071
Shapiro-Wilk p-value	2.43e-8
CV error	1.84623

SE + t

Question 11: State the estimated value of β_1, and interpret its value in the context of the problem

For every additional inch, the hot dog eaters can eat .472 additional hot dogs on average in 10 minutes.

$\beta_1 = .4729$

Question 12: **(Circle all that apply)** Which of the following are true?

(A) The 95% confidence interval of β_2 does **not** contain 0

(B) The 95% confidence interval of β_2 does contain 0

(C) There is an 81.4% probability that $\beta_0 = 0$

(D) $\beta_1 = 0.4729$

Question 13: Given what you know, provide a brief assessment of how well the data satisfy each of the following multiple linear regression assumptions:

(A) The relationships between the explanatory and response variables are linear

Residuals v fitted shows spread out scattering. Added variable plots may have some curving for minutes, but otherwise it is probably fine.

(B) There is independence between each observation

Because the data was collected from 1 contest at 1 state fair, I would argue there is a high chance they are not dependent.

(C) The errors are normally distributed

The Q-Q plot shows some skewness. Additionally, the shapiro wilk p value indicates it is abnormal. I would definitely be concerned about this.

(D) The errors have constant variance (homoscedasticity)

The residuals vs fitted doesn't seem to have significant "funneling". The BP test also indicates equal variance.

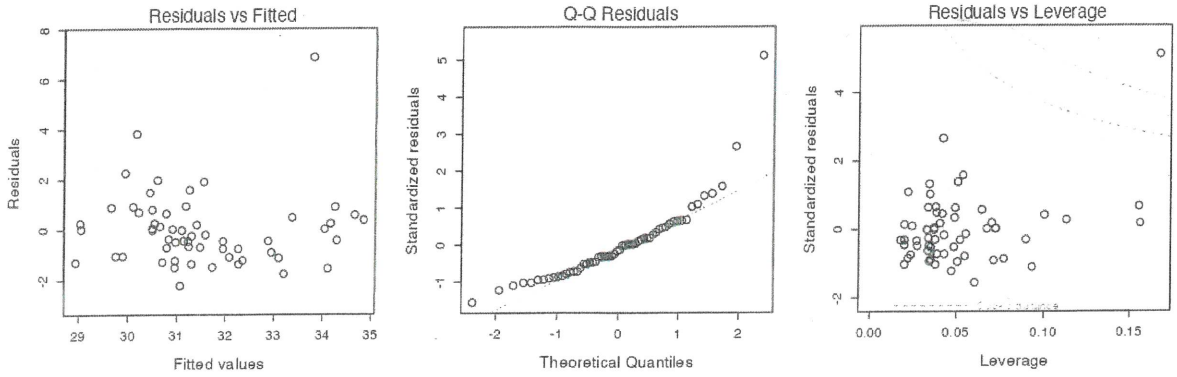
(E) There are no unduly influential points

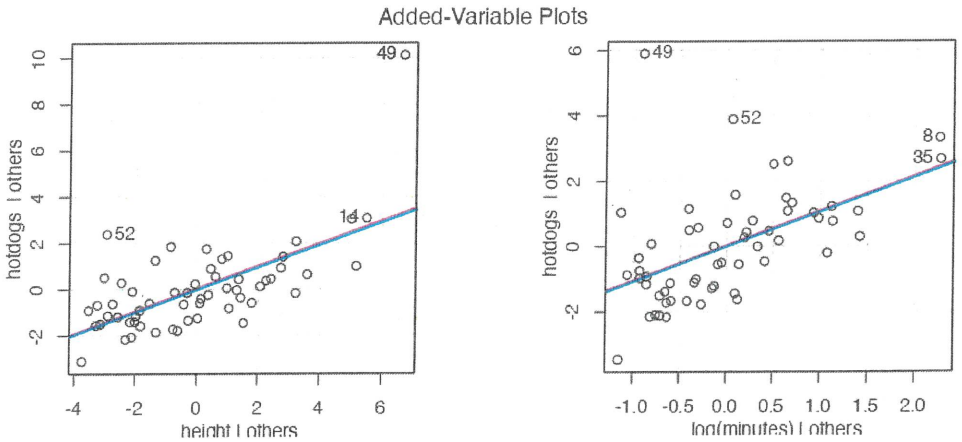
There appears to be one influential point in the residuals vs leverage plot. There are some points with high leverage as well.

Model 2

Model (2) below represents another possible multiple linear regression model for the same data. Here the variable `minutes` was log-transformed prior to fitting the model. Use model (2) and the output below to answer questions 14-16

$$\texttt{hotdogs}_i = \beta_0 + \beta_1 \texttt{height}_i + \beta_2 \log(\texttt{minutes}_i) + \epsilon_i; \quad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2) \tag{2}$$





Coefficient	Estimate	Std. Error	t value	$\Pr(>\|t\|)$
Intercept	-6.05589	5.43620	-1.114	0.27
`height`	0.48834	0.07821	6.244	6.48e-08
log(`minutes`)	1.06702	0.24079	4.431	4.51e-05

Other Summaries	Value
Breusch-Pagan p-value	0.028
Shapiro-Wilk p-value	3.43e-5
CV RMSE	1.66702

Question 14: A Breusch-Pagan test for model (1) results in a p-value of 0.071 while a Breusch-Pagan test for model (2) results in a p-value of 0.028. A researcher concludes that model (1) is clearly superior to model (2). How would you respond?

While the BP test indicates less unequal variance, the fitted vs residuals does not seem to have less funneling. It is important to consider other aspects of the model.

Question 15: Explain what advantage(s) and disadvantage(s) model (2) provides over model (1)

model 2: has a lower CV RMSE
has lower p values
lower BP test
similar shapiro wilk
} advantages

some points are more influential - disadvantage

Question 16: State the estimated value of β_2 and provide an interpretation in context

(c)
A 10% increase in the amount of minutes spent practicing is associated with an average increase of .04 hot dogs eaten.

$\log(1 + c/100)$

Use the following for questions 17-19

Your best friend reveals that he will be competing in the hotdog eating contest at the next state fair. You know that he is 68 inches tall and that he trains 79 minutes per week. Note that log(79) = 4.369. Use this information and the fit for model (2) to answer the following questions.

Question 17: Use model (2) to calculate the predicted number of hotdogs your friend can consume in 10 minutes

$68(.488) + 4.369(1.067) =$

$33.184 + 4.66 = 37.845$ hot dogs

Question 18: To create a confidence or prediction interval for the expected number of hotdogs your friend can consume in 10 minutes, what is the correct number of degrees of freedom for the critical t^* used to calculate the margin of error?

(A) 55

(B) 57

(C) 56

(D) 48

$n - p - 1$

$58 - 2 - 1$

Question 19: You're given that $s\sqrt{x^*(X^\top X)^{-1}x^*} = 0.204$, $s\sqrt{1 + x^*(X^\top X)^{-1}x^*} = 1.485$, and $t^*_{df,0.975} = 2.004$. Give a 95% *prediction* interval for the number of hotdogs your friend can consume in 10 minutes.

accounts for prediction variability

$y \pm 2.004(1.485)$

37.845 ± 2.975

34.87 - 40.82

Part III - Application

Questions 20-24 concern the following dataset:

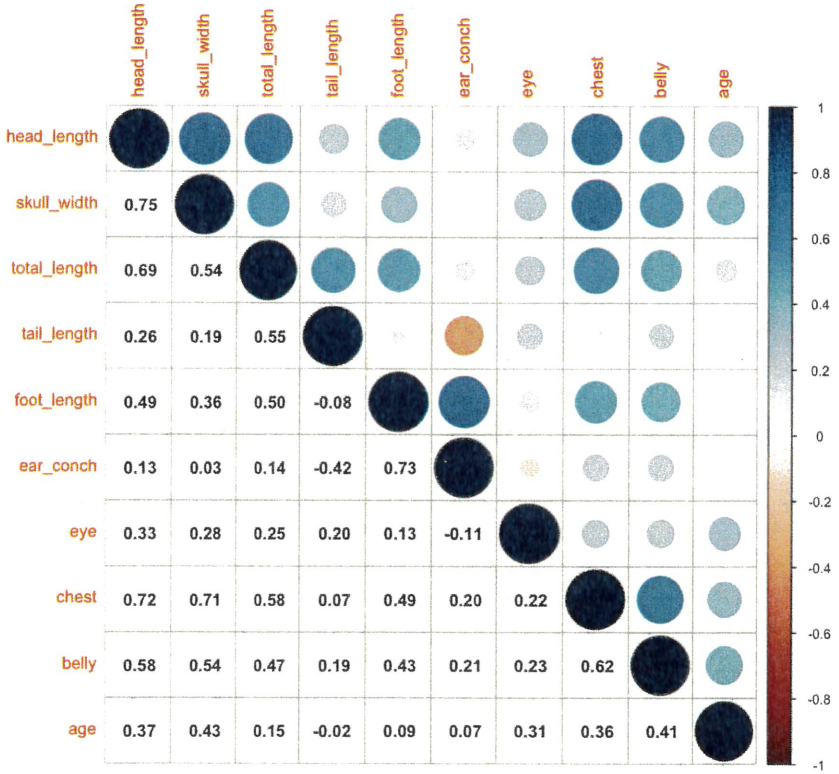
In 1995, Australian zoologists conducted a study of local possum populations. They trapped dozens of specimens, determined their ages, and took measurements of several physical attributes from each possum. The process of determining a wild animal's age can be challenging and imprecise, and usually requires a detailed analysis of the animal's teeth, skin, skeletal structure, or DNA. Among other questions, researchers were interested in whether a possum's age could be predicted using only easy to obtain physical measurements. For this analysis we will focus on the 59 male possums from the sample. The variables in the dataset are as follows (see below). A multiple linear regression model to predict age using all other variables was fit to these data, resulting in the following model summary and coefficient estimates (see below)

Variable	Description
head_length	in mm
skull_width	in mm
total_length	in cm
tail_length	in cm
foot_length	in mm
ear_conch	ear conch circumference (in mm)
eye	width of right eye (in mm)
chest	chest girth (in cm)
belly	belly girth (in cm)
age	in years

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.22377    7.34105  -1.801   0.0778 .
head_length   0.07077    0.10120   0.699   0.4877
skull_width   0.16559    0.10599   1.562   0.1247
total_length -0.04752    0.10167  -0.467   0.6423
tail_length  -0.05049    0.17030  -0.296   0.7681
foot_length  -0.21499    0.10733  -2.003   0.0507 .
ear_conch     0.15730    0.10231   1.538   0.1306
eye           0.43736    0.22549   1.940   0.0582 .
chest         0.01883    0.18154   0.104   0.9178
belly         0.20983    0.11580   1.812   0.0761 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.655 on 49 degrees of freedom
Multiple R-squared:  0.3538,    Adjusted R-squared:  0.2352
F-statistic: 2.982 on 9 and 49 DF,  p-value: 0.006552
```

A correlation plot (shown below) of these variables points towards some level of multicollinearity in the data. The maximum VIF among these variables is approximately 4 with an average VIF of approximately 3.



Question 20:

(a) Using the information from the summary table above, produce a 95% confidence interval for the coefficient on `skull_width`. The 0.975 critical value of a t-distribution with 49 degrees of freedom is 2.01. You may report your answer in the form $a \pm b * c$.

coef •
y = .16959

.16959 ± 2.01 × .10599

(b) A 95% confidence interval for $\beta_{\texttt{belly}}$, the coefficient on belly width, is (-0.023, 0.443). Interpret this interval within the context of the data.

Because the interval includes 0, we can't conclude the belly width is an influential measure of a possum's age.

Question 21: Note from the model summary that while the overall F-test of model significance produces a p-value of 0.0065, none of the individual model coefficients are found to have a significant effect at the $\alpha = 0.05$ level. What do you think is driving this "contradiction"? (It's not actually a contradiction.)

Question 22: Briefly describe how the "best subsets" algorithm using AIC as its criterion performs variable selection.

The AIC selection metric prioritizes variable selection for models with lower cross validation error. This is good for prediction, or if you want a larger model than BIC would offer.

Question 23: Best subsets using AIC as a criterion identified the following model with five variables:
`lm(age ~ skull_width + foot_length + ear_conch + eye + belly, data = possum)`
This model produces the following summary and coefficient estimates:

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -14.76661    4.97828  -2.966  0.00451 **
skull_width   0.20221    0.07990   2.531  0.01439 *
foot_length  -0.22936    0.09385  -2.444  0.01789 *
ear_conch     0.18239    0.08653   2.108  0.03979 *
eye           0.45001    0.21687   2.075  0.04285 *
belly         0.21405    0.10316   2.075  0.04287 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.609 on 53 degrees of freedom
Multiple R-squared:  0.3396,    Adjusted R-squared:  0.2773
F-statistic:  5.45 on 5 and 53 DF,  p-value: 0.0004056
```

Is this model guaranteed to have the highest adjusted-R^2 among all possible models.

(A) TRUE

(B) FALSE

Question 24: Suppose we want to compare the out-of-sample predictive performance of the full model and the model chosen by best subsets, and ultimately select the model with the smaller cross-validated RMSE.

Cross-validation using repeated random training-test splits can be structured around a for loop over some number of iterations (let's say 100), with some setup happening prior to the start of the loop, the bulk of the code within the loop, and some code after the loop has ended to process the results. To be clear, we **ARE NOT** using k-fold cross validation.

For each of the following steps in the cross-validation process, identify (by writing in the blank space) whether it occurs:

(A) prior to the for loop

(B) within the for loop

(C) after the for loop

(D) never in the CV process

C Average RMSE across all test-training splits and report

D Average the predictions and record their value

D Choose the model with the largest RMSE

C Choose the model with the smallest RMSE

B Compare the predictions from each model (that was fitted to training data) to the observed test data and compute RMSE

C Compute each model's AIC and BIC

D Fit each model using the testing data

B Fit each model using the training data

A Initialize vectors for each model that you will use to store their RMSEs

D Refit each model using the full data set and compare the coefficient estimates to the true values

B Split the data into test and training sets

B Use the models fitted to the training data to get predictions for the test data

END OF EXAM